CAT NIPS ENGINE

PRESENTATION

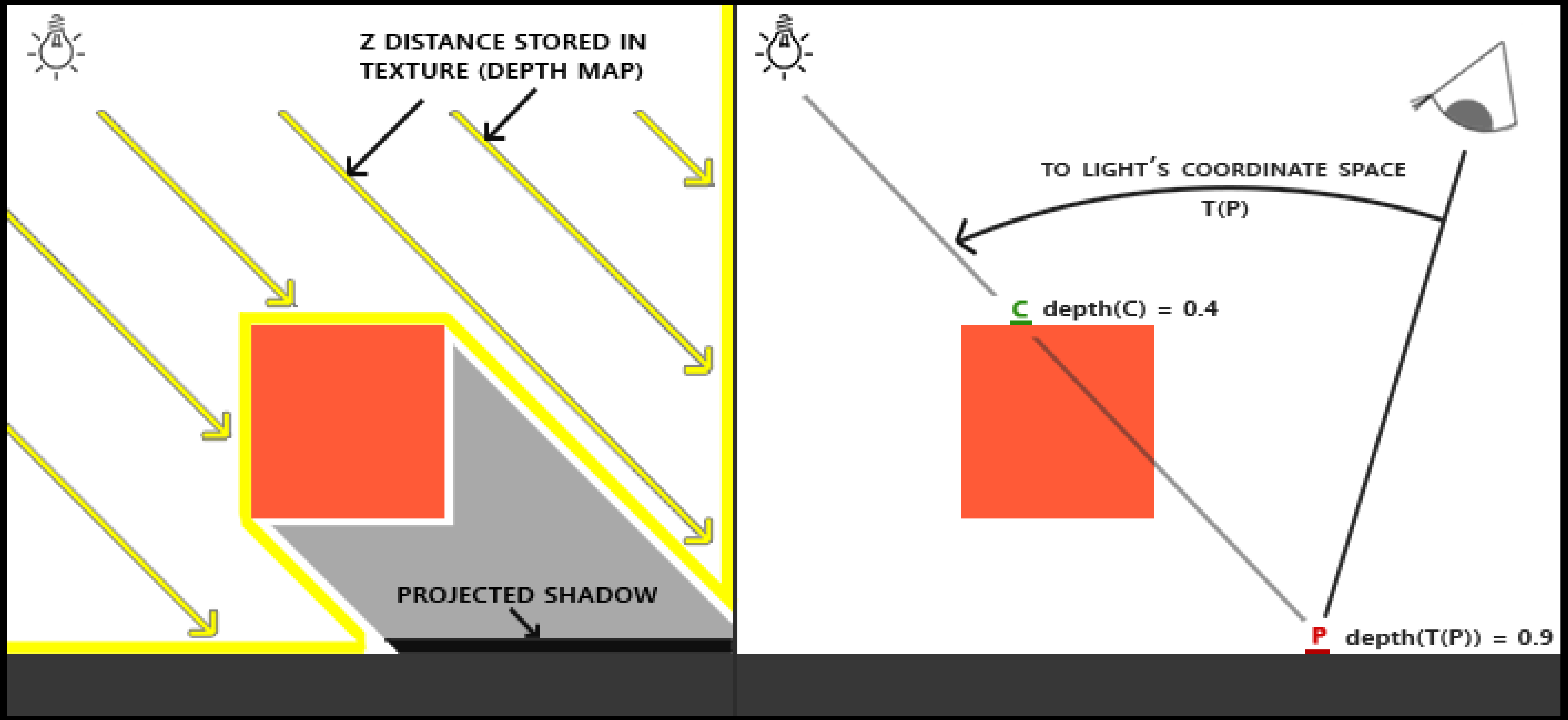
JUNGLE *GAME TECH LAB* 3 W8
TEAM1 김지수 김형도 권현수

발제 내용

- Standard Shadow Mapping
- Shadow Mapping Artifacts & Solutions
- Shadow Atlas

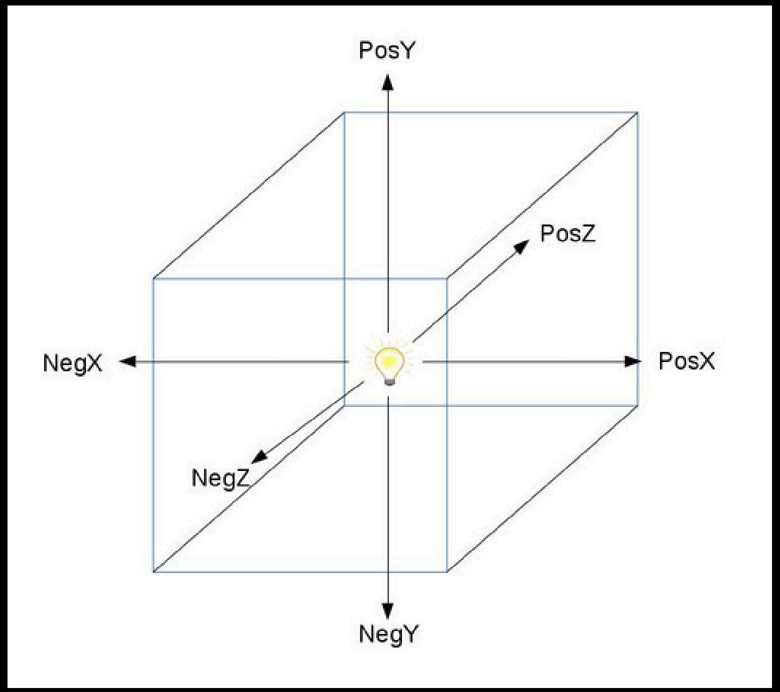
GAME TECH LAB TEAM 1

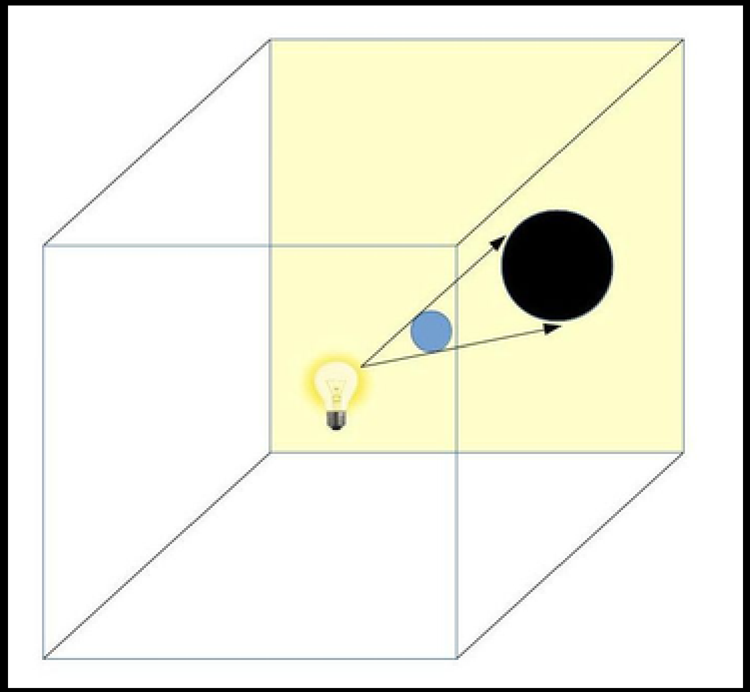
Standard Shadow Mapping (SSM)

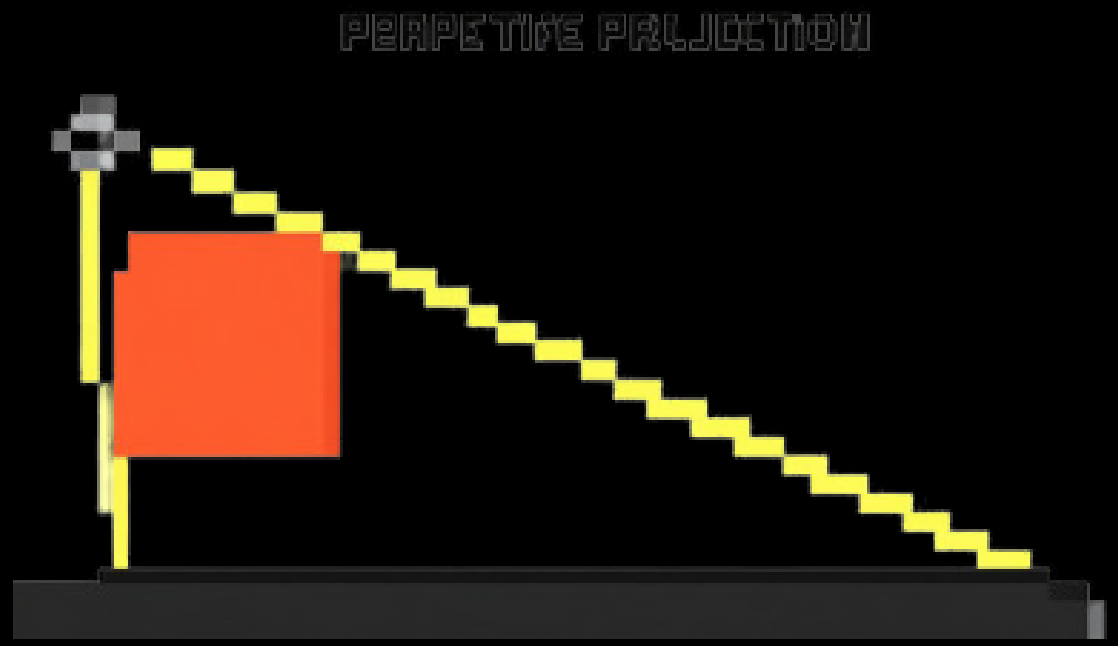



```
Shadow = (SampleDepth >= CurrentDepth) ? 1.0f : 0.0f;
```

Standard Shadow Mapping (SSM)

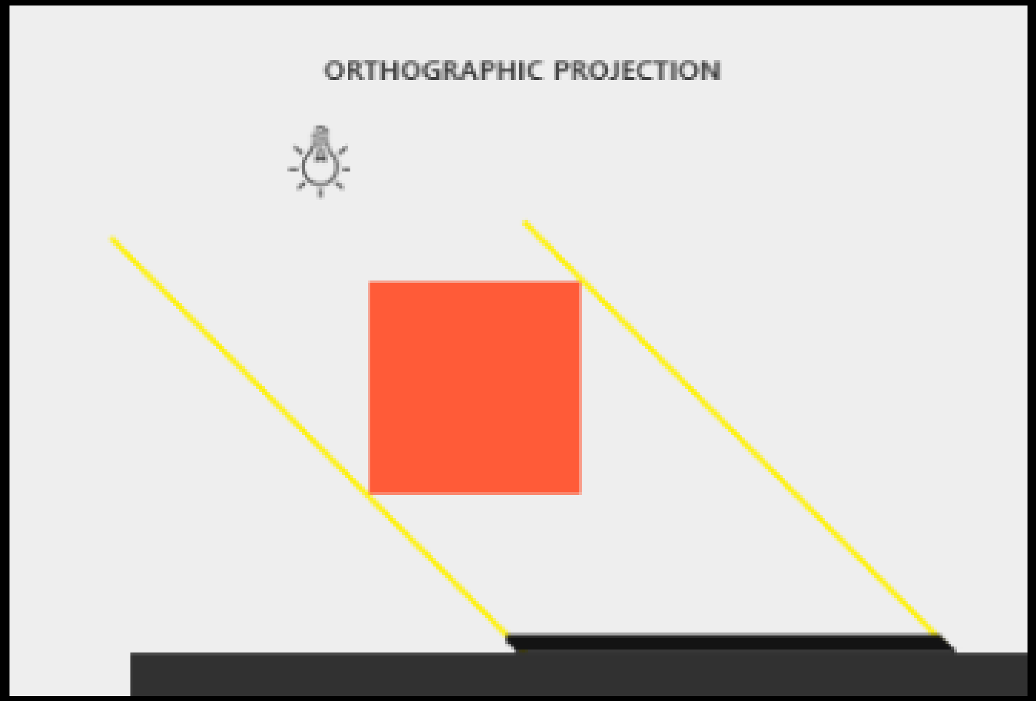


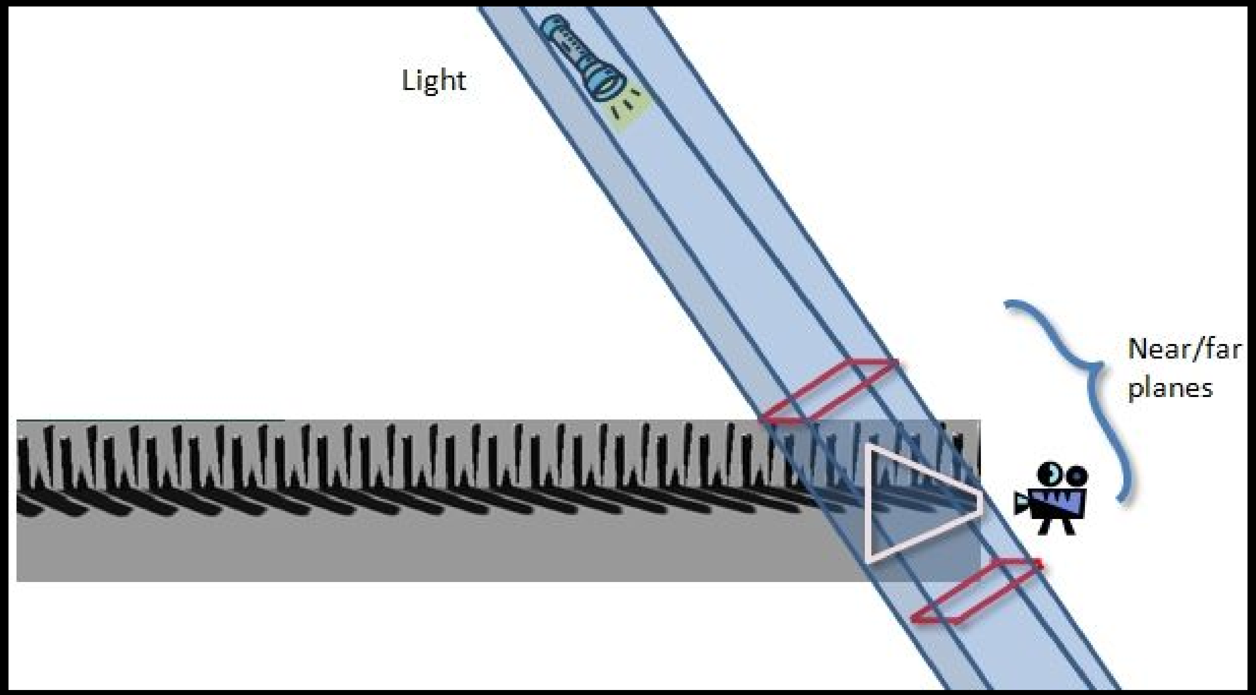




Point Light / Spot Light

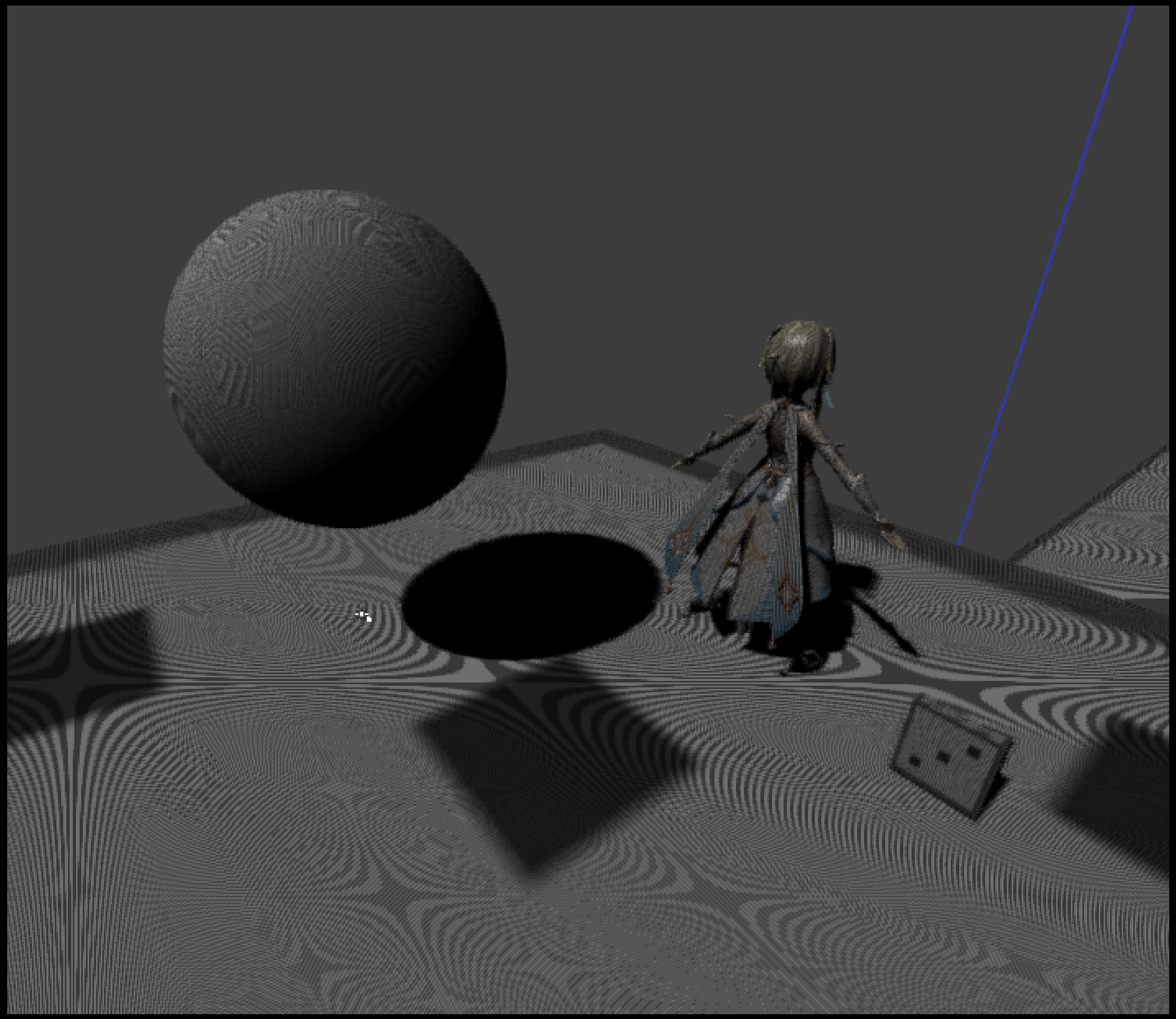
Standard Shadow Mapping (SSM)

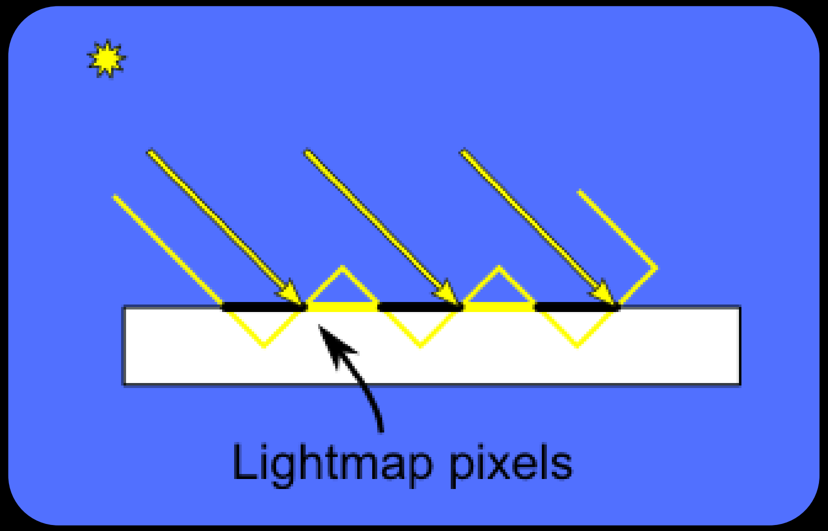




Directional Light

Shadow Acne





그림자가 있는 표면에
줄무늬 같은 아티팩트가 나타나는 현상

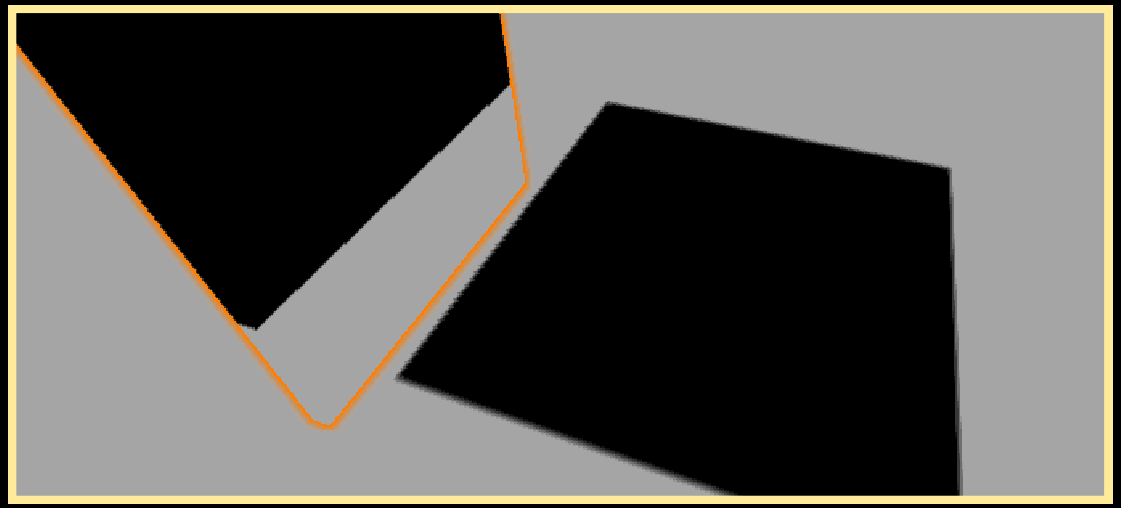
Depth Bias

1. Constant Bias

```
float bias = 0.01;
Shadow += (SampleDepth + bias >= CurrentDepth) ? 1.0f : 0.0f;
```

↓

2. Peter Panning



↓

3. Slope Bias

```
float CosTheta = saturate(dot(N, L));
float Slope = sqrt(1 - CosTheta * CosTheta) / max(CosTheta, 1e-4);
Shadow += (SampleDepth + Slope >= CurrentDepth) ? 1.0f : 0.0f;
```

Shadow Map Aliasing

Shadow Map의 해상도가
화면 해상도에 비해 부족할 때 발생하는 현상

Percentage Closer Filtering (PCF)

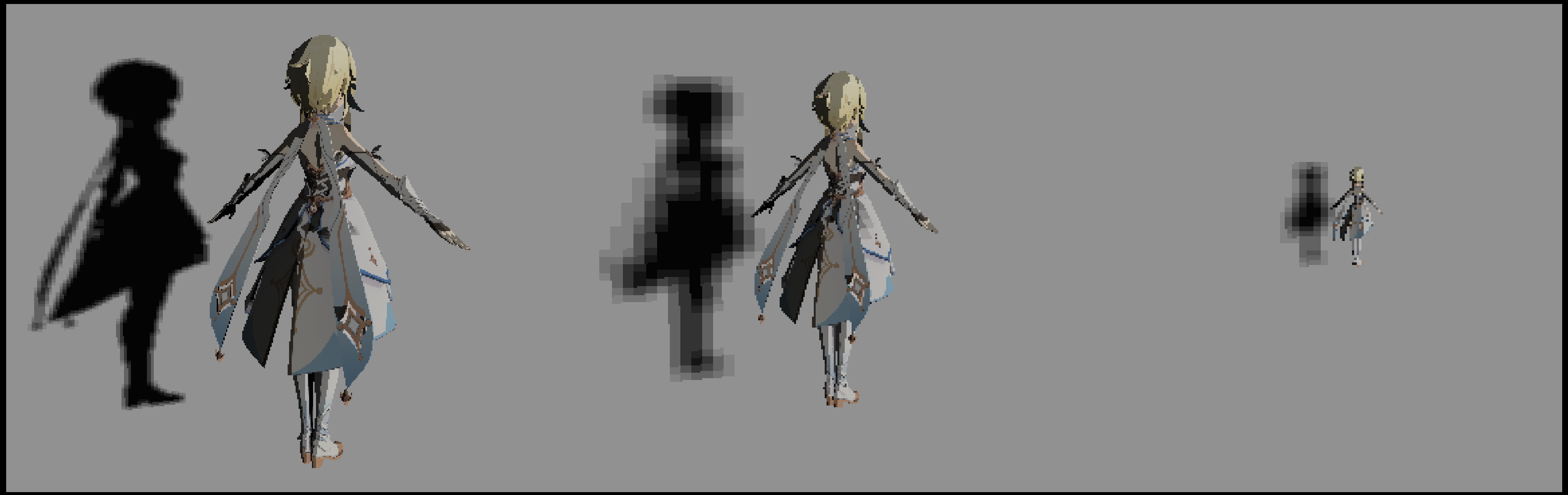
...	...	...	...	...
...	0	0	1	...
...	0	1	1	...
...	1	1	1	...
...	...	...	...	...

Sampling an Area of 3x3 Texels

```
ShadowMask = (0 * 3 + 1 * 6) / 9;
    ...
Lighting.Diffuse += LightContribution * NdotL * ShadowMask;
Lighting.Specular += SpecularColor * LightContribution
                                       * SpecularPower * ShadowMask;
    ...
return Surface.Albedo * Lighting.Diffuse + Lighting.Specular;
```

Percentage Closer Filtering (PCF)

Cascading Shadow Map (CSM)



Cascade 0　　Cascade 1　　Cascade 2

카메라 View Frustum을 여러 구간으로 나누고
각 구간마다 **별도의 Shadow Map을 생성**하는 기법

Variance Shadow Map (VSM)

…	…	…	…	…
…	d : 0.5 d^2 : 0.25	d : 0.6 d^2 : 0.36	d : 0.8 d^2 : 0.64	…
…	d : 0.6 d^2 : 0.36	d : 0.8 d^2 : 0.64	d : 0.7 d^2 : 0.49	…
…	d : 0.3 d^2 : 0.09	d : 0.3 d^2 : 0.09	d : 0.3 d^2 : 0.09	…
…	…	…	…	…

VSM Moment Map의 raw texel 예시 (blur 전)

< 중심 texel의 3×3 Box Blur 계산 >

$$M_1 = \frac{0.5+0.6+0.8+0.6+0.8+0.7+0.3+0.3+0.3}{9} = \frac{4.9}{9} \approx 0.5444$$

$$M_2 = \frac{0.25+0.36+0.64+0.36+0.64+0.49+0.09+0.09+0.09}{9} = \frac{3.01}{9} \approx 0.3344$$

$$Var = M_2 - M_1^2 = 0.3344 - 0.2964 \approx 0.0380$$

Variance Shadow Map (VSM)

d : 0.5 d^2 : 0.25	d : 0.6 d^2 : 0.36	d : 0.8 d^2 : 0.64
d : 0.6 d^2 : 0.36	d : 0.8 d^2 : 0.64	d : 0.7 d^2 : 0.49
d : 0.3 d^2 : 0.09	d : 0.3 d^2 : 0.09	d : 0.3 d^2 : 0.09

blur 전: Raw Moment

각 texel = (d, d^2)

$\longrightarrow$

3x3

Box Blur

M_1 : ... M_2 : ... $Var(\sigma^2)$: ...	M_1 : ... M_2 : ... $Var(\sigma^2)$: ...	M_1 : ... M_2 : ... $Var(\sigma^2)$: ...
M_1 : ... M_2 : ... $Var(\sigma^2)$: ...	M_1 : 0.544 M_2 : 0.334 $Var(\sigma^2)$: 0.038	M_1 : ... M_2 : ... $Var(\sigma^2)$: ...
M_1 : ... M_2 : ... $Var(\sigma^2)$: ...	M_1 : ... M_2 : ... $Var(\sigma^2)$: ...	M_1 : ... M_2 : ... $Var(\sigma^2)$: ...

blur 후: Filtered Moment

각 texel = (M_1, M_2), $Var(\sigma^2)$

VSM in UberLit: Visibility 계산

Blurred VSM Texel

M_1 : 0.544
M_2 : 0.334
$Var(\sigma^2)$: 0.038

현재 픽셀의 light-space depth = t

$t \le M_1$

→ visibility = 1.0

→ 그림자 아님

$t > M_1$

→ $pmax = \sigma^2 / (\sigma^2 + (t - M_1)^2)$

→ visibility ≈ pmax

UberLit에서 Shadow 계산 값으로 visibility를 사용.

VSM in UberLit: Visibility 계산

Blurred VSM Texel

M_1 : 0.544
M_2 : 0.334
$Var(\sigma^2)$: 0.038

< Case A >

t = receiverDepth = 0.40

$t \le M_1$
$0.40 \le 0.544$

visibility = 1.0
→ 그림자 아님

< Case B >

t = receiverDepth = 0.80

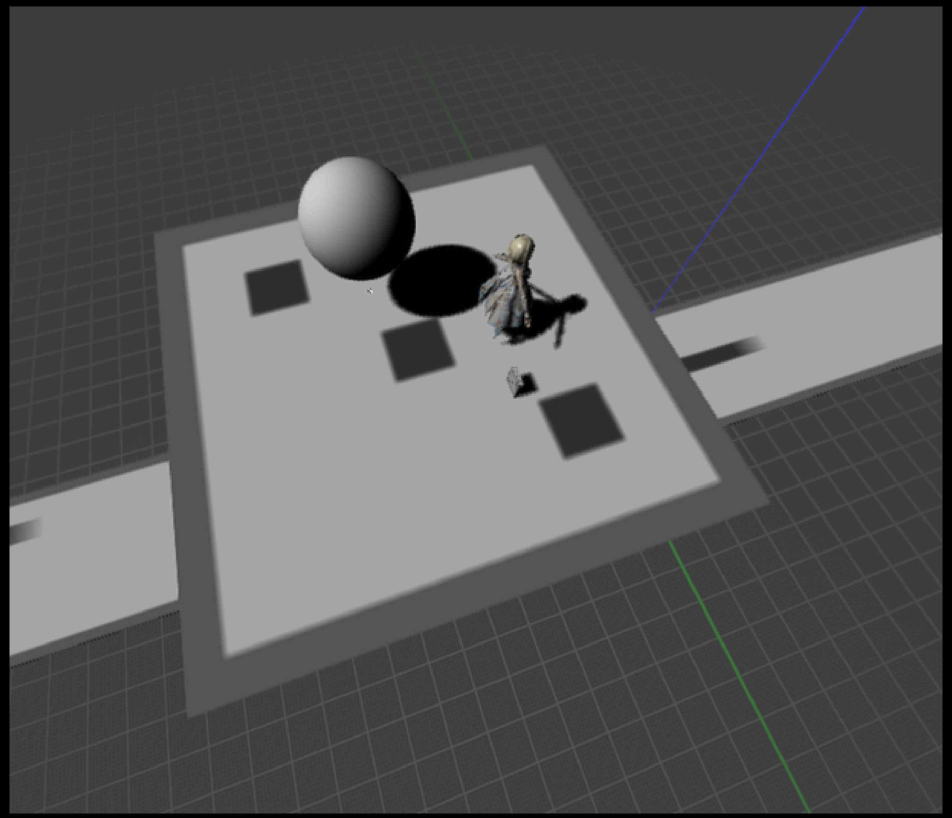
$t > M_1$
$0.80 > 0.544$

$d = t - M_1 = 0.256$

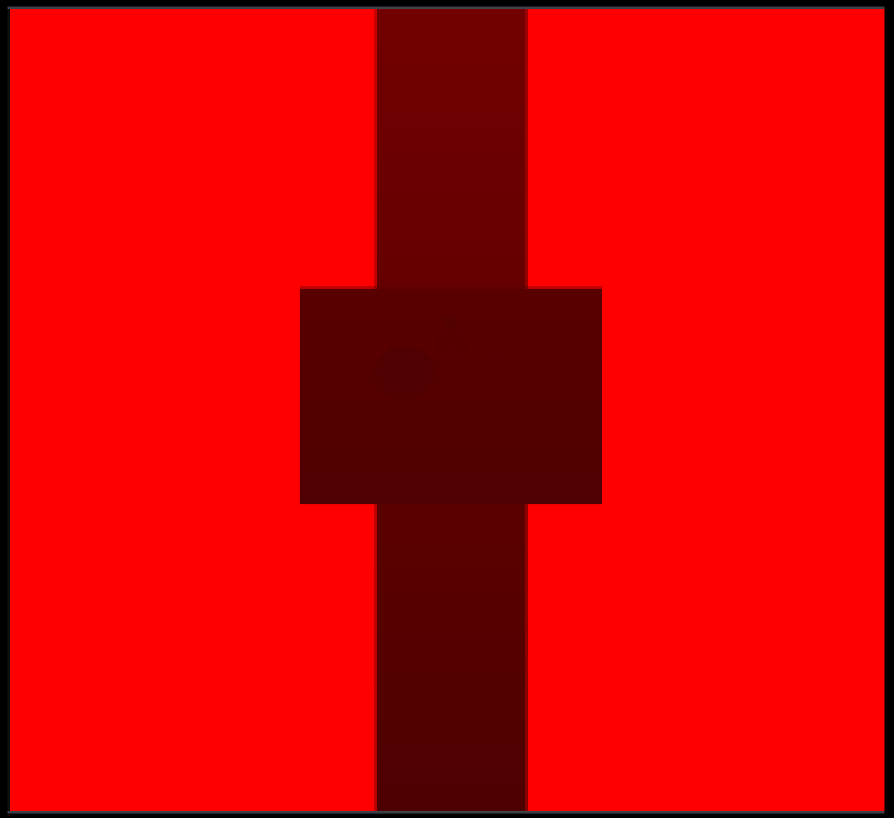
$pmax = 0.038 \,/\, (0.038 + 0.256^2)$
≈ 0.367

visibility ≈ 0.367
→ direct light의 36.7%만 적용

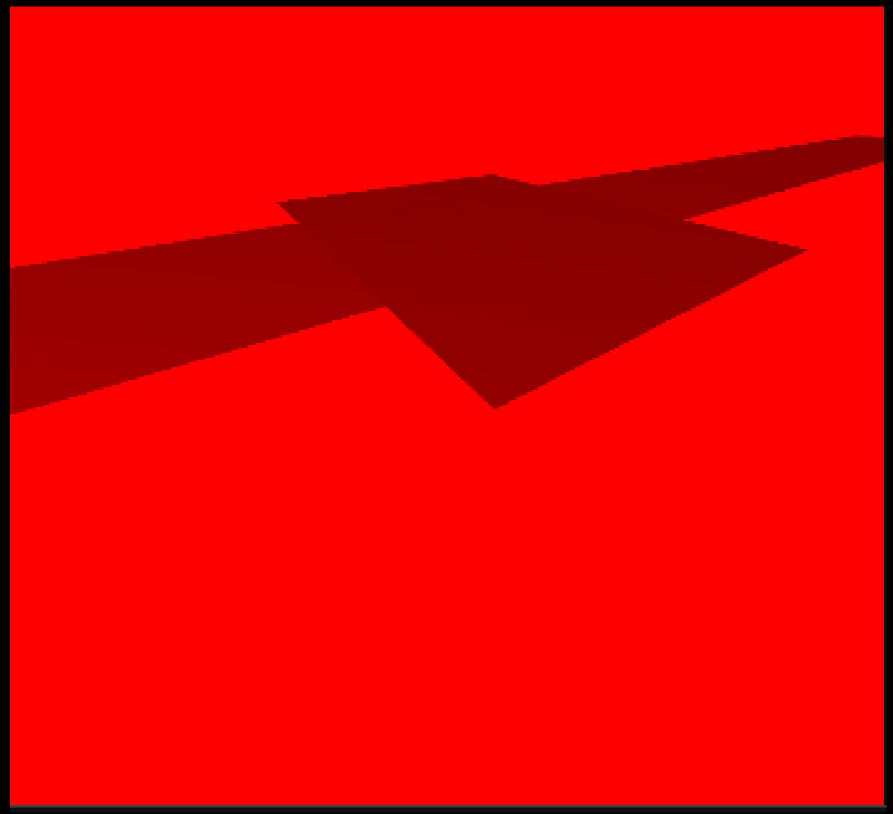
Perspective Shadow Map (PSM)



카메라 시점

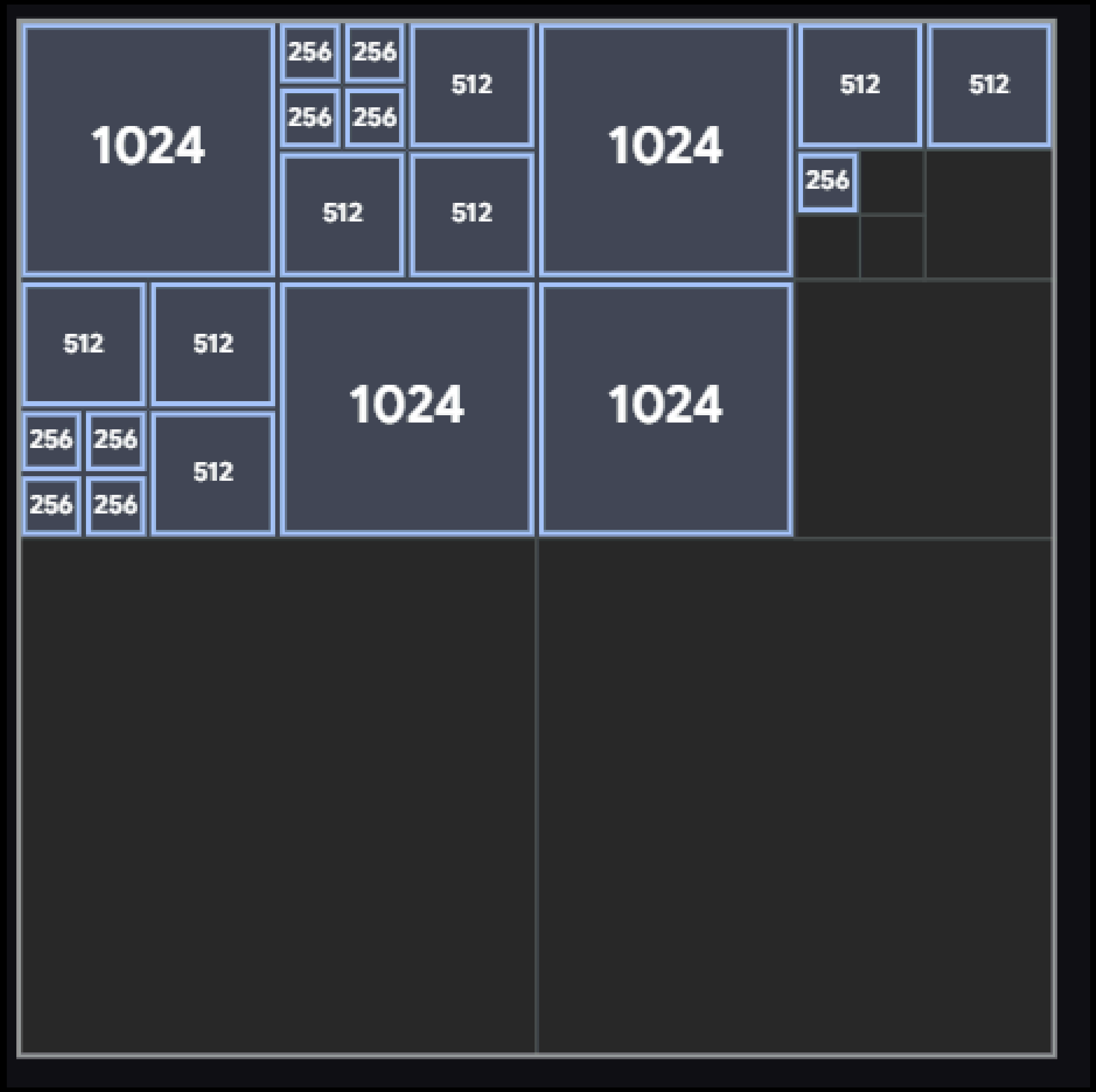


uniform shadow map depth



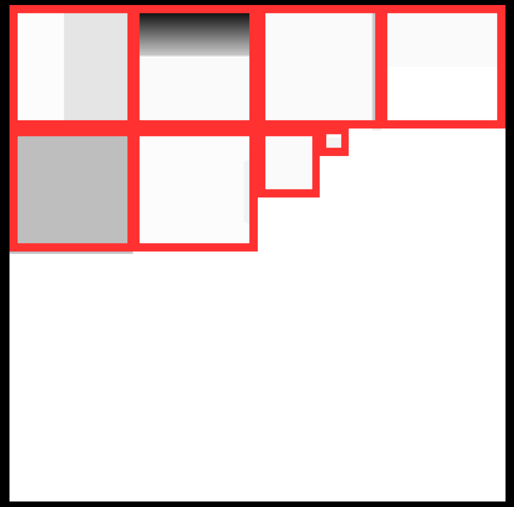
perspective shadow map depth

Shadow Atlas



Quad Tree 사용

2차원 공간을 재귀적으로
네 개의 사분면으로 분할



시연